

3(a)	<i>do not allow 'base' as AW for 'nucleotide'</i> any one from: <i>idea that phosphodiester bond forms,</i> between nucleotides on the same strand / between adjacent nucleotides / during polynucleotide formation / to form a sugar-phosphate backbone ; AW <i>look for understanding that more than one nucleotide needs to be present on the same strand</i> <i>idea that phosphates, shown are only bound to one (pentose) sugars / each phosphate needs to be bound to two (pentose) sugars ;</i>	1
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Question	Answer	Marks
3(b)	base / nucleotide, on left identified as <u>guanine</u> <u>and</u>, base / nucleotide, on right identified as <u>cytosine</u> ; A guanine and cytosine as question has DNA-RNA I cytosine and guanine unless further qualified dotted line as hydrogen bond / annotation related to dotted lines as hydrogen bonds / there are three hydrogen bonds between the bases; A <u>H</u>-bond circle is phosphate ; pentagon / pentose, on left is deoxyribose ; also AVP if 'pentose' double ring / base on left, identified as purine ; A guanine is a purine if mp1 gained single ring / base on right, identified as pyrimidine ; A cytosine is a pyrimidine if mp1 gained AVP ; e.g. label to any solid line bond as covalent bond phospho-ester bond, labelled / described pentose sugar, label to / described for, deoxyribose / ribose box drawn around one completed nucleotide and labelled 5' and 3' shown annotated ref. to antiparallel nature of base-pair	4